

README for Replication Package

Municipality Proliferation

June 22, 2026

Overview

This replication package contains the Stata and R code used to construct the analysis data and reproduce the tables and figures for the municipality proliferation project. The main entry point is `code/master.do`. That file sets all paths, writes `paths.csv` for R scripts, runs the Derenoncourt replication setup used by this project, constructs cleaned/interim datasets, and then runs the analysis scripts that write tables to `exhibits/tables` and figures to `exhibits/figures`.

The current code is organized around an external data directory rather than a `data/` directory inside the Git repository. In `code/master.do`, `$DROPBOX` is currently set to `D:/munis_replication/` for user `EverettStamm` and `F:/munis_replication/` for user `edog9`. The code then expects:

- `$DROPBOX/data/raw` for raw inputs.
- `$DROPBOX/data/interim` for generated intermediate files.
- `$DROPBOX/data/clean` for final cleaned analysis files.
- `$DROPBOX/data/xwalks` for crosswalk inputs and generated crosswalks.

The exact default data root at the time of this draft is `D:/munis_replication/data`. A replicator should edit the path block at the top of `code/master.do` before running the code.

TODO: Before deposit, confirm whether the final package will keep data outside the repository as in the current code, or whether the data will be deposited under a package-level `data/` directory. If the latter, update `code/master.do` so path changes are limited to one portable configuration block.

Data Availability and Provenance Statements

Statement About Rights

- ☐ I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.
- ☐ I certify that the author(s) of the manuscript have documented permission to redistribute/publish the data contained within this replication package.

TODO: Check these boxes only after confirming the redistribution terms for all raw and derived inputs, especially IPUMS extracts, Derenoncourt replication inputs, CoreLogic-derived files, and any court-order or school-achievement files.

License for Data

TODO: Specify the data license for the replication package. The repository contains a `LICENSE` file, but this README draft does not determine whether it covers code only or both code and

redistributed data. Some source data may have inherited licenses or redistribution restrictions.

Summary of Availability

- ☐ All data are publicly available.
- ☐ Some data cannot be made publicly available.
- ☐ No data can be made publicly available.

Based on the code inspection, the package appears to rely on a mixture of public-use data, public data with registration or citation requirements, pre-generated intermediate data, and possibly restricted/proprietary data. The current repository does not contain the `data/` directory used by `code/master.do`.

TODO: Select the correct availability statement after confirming whether the final data deposit will include all raw and derived data, or whether some data must be omitted or provided separately.

Details on Each Data Source

The table below lists the external and pre-generated data inputs identified from programs called by `code/master.do`. The Derenoncourt replication package is intentionally treated as one upstream input, as requested. Census, NHGIS, Census of Governments, and Individual Finance files are split into separate rows so source checks can be completed one file family at a time.

Data source	Files or folders used by code	Expected location	Use and availability notes
Derenoncourt replication package	<code>dcourt/replication_AER/</code> ; includes Derenoncourt migration, crosswalk, city population, industry employment, and mechanism files used throughout the pipeline.	<code>data/raw/dcourt/replication_AER/</code> ; outputs also written to <code>data/interim/dcourt/</code> .	Treat as a single input. TODO: Add exact citation, DOI/URL, license, and redistribution status.
C. Goodman municipal incorporation data	<code>cbgoodman/muni_incorporation_date.dta</code>	<code>data/raw/cbgoodman/</code>	Used to construct municipal incorporation outcomes and shapefile attributes. TODO: Confirm source, citation, and permission to redistribute.
IPUMS USA extract	<code>census/usa_00072.dta/usa_00072.dta</code>	<code>data/raw/census/</code>	Used by <code>black_linked_characteristics.do</code> for linked Black migrant characteristics. TODO: Confirm extract definition, citation, and redistribution rules.
IPUMS USA extract	<code>census/usa_00108.dta/usa_00108.dta</code>	<code>data/raw/census/</code>	Used by <code>cz_pop_occscore_mfg.do</code> for full-count CZ population, race, OCCSCORE, and manufacturing share variables. TODO: Confirm year/content of this extract.
IPUMS USA extract	<code>census/usa_00109.dta/usa_00109.dta</code>	<code>data/raw/census/</code>	Used by <code>cz_pop_occscore_mfg.do</code> . TODO: Confirm year/content of this extract.
IPUMS USA extract	<code>census/usa_00110.dta/usa_00110.dta</code>	<code>data/raw/census/</code>	Used by <code>cz_pop_occscore_mfg.do</code> . TODO: Confirm year/content of this extract.
IPUMS USA extract	<code>census/usa_00111.dta/usa_00111.dta</code>	<code>data/raw/census/</code>	Used by <code>cz_pop_occscore_mfg.do</code> . TODO: Confirm year/content of this extract.
IPUMS USA extract	<code>census/usa_00116.dta/usa_00116.dta</code>	<code>data/raw/census/</code>	Used by <code>cz_pop_occscore_mfg.do</code> . TODO: Confirm year/content of this extract.
IPUMS USA extract	<code>census/usa_00118.dta/usa_00118.dta</code>	<code>data/raw/census/</code>	Used by <code>incomes.do</code> , <code>education.do</code> , and <code>cz_pop_occscore_mfg.do</code> for 1940 income, education, and full-count CZ variables. TODO: Confirm extract definition and IPUMS citation.

Data source	Files or folders used by code	Expected location	Use and availability notes
Census Gazetteer / national county file	census/national_county.txt	data/raw/census/	Used by <code>place_county_xwalk.do</code> to build the place-county crosswalk. TODO: Add exact Census citation and access date.
Census Gazetteer / national places file	census/national_places.txt	data/raw/census/	Used by <code>place_county_xwalk.do</code> to build the place-county crosswalk. TODO: Add exact Census citation and access date.
Census metro reference file	census/99mfips.txt	data/raw/census/	Used by <code>cog_cleaning.do</code> for PMSA/MSA names and county-PMSA crosswalks. Source comment points to the 1999 Census historical delineation file.
Census place FIPS crosswalk	census/census_place_fips_xwalk.txt	data/raw/census/	Used by <code>incomes.do</code> and <code>mechanisms.do</code> to translate 1970 Census place codes to FIPS place codes. TODO: Confirm source and citation.
Geocorr block-group to place crosswalk	census/geocorr/geocorr2014_mo_il.csv	data/raw/census/geocorr/	Used by <code>incomes.do</code> for 2010 place-level household income. TODO: Confirm Geocorr version, parameters, and citation.
Geocorr block-group to place crosswalk	census/geocorr/geocorr2014_in_ne.csv	data/raw/census/geocorr/	Used by <code>incomes.do</code> . TODO: Confirm Geocorr version, parameters, and citation.
Geocorr block-group to place crosswalk	census/geocorr/geocorr2014_nv_sc.csv	data/raw/census/geocorr/	Used by <code>incomes.do</code> . TODO: Confirm Geocorr version, parameters, and citation.
Geocorr block-group to place crosswalk	census/geocorr/geocorr2014_sd_wy.csv	data/raw/census/geocorr/	Used by <code>incomes.do</code> . TODO: Confirm Geocorr version, parameters, and citation.
NHGIS extract	census/nhgis0017_csv/nhgis0017_csv/nhgis0017_ts_nominal_county.csv	data/raw/census/	Used by <code>cz_place_race_pop_1970_2010.do</code> for county race/population time series. TODO: Add NHGIS extract citation and access date.
NHGIS extract	census/nhgis0019_csv/nhgis0019_ds94_1970_county.csv	data/raw/census/	Used by <code>cz_place_race_pop_1970_2010.do</code> for 1970 county race/population. TODO: Add NHGIS extract citation.

Data source	Files or folders used by code	Expected location	Use and availability notes
NHGIS extract	census/nhgis0020_fixed/nhgis0020_fixed/nhgis0020_ds172_2010_county.do and associated data files	data/raw/census/	Used by <code>cz_pop_occscore_mfg.do</code> for 2010 urban population. TODO: Confirm all files required by the NHGIS Stata import script.
NHGIS extract	census/nhgis0025_csv/nhgis0025_csv/nhgis0025_ds258_2020_place.csv	data/raw/census/	Used by <code>municipal_shapefile.R</code> for 2020 place population attributes. TODO: Add NHGIS extract citation.
NHGIS extract	census/nhgis0027_csv/nhgis0027_csv/nhgis0027_ts_national_place.csv	data/raw/census/	Used by <code>cz_place_race_pop_1970_2010.do</code> for 1970 and 2010 place race/population variables. TODO: Add NHGIS extract citation.
NHGIS extract	census/nhgis0035_csv/nhgis0035_csv/nhgis0035_ds176_20105_blk_grp.csv	data/raw/census/	Used by <code>incomes.do</code> for 2010 household income at CZ and place level. TODO: Add NHGIS extract citation.
NHGIS extract	census/nhgis0036_csv/nhgis0036_csv/nhgis0036_ds99_1970_county.csv	data/raw/census/	Used by <code>incomes.do</code> for 1970 CZ family income. TODO: Add NHGIS extract citation.
NHGIS extract	census/nhgis0036_csv/nhgis0036_csv/nhgis0036_ds99_1970_place.csv	data/raw/census/	Used by <code>incomes.do</code> for 1970 place family income and house value variables. TODO: Add NHGIS extract citation.
NHGIS extract	census/nhgis0038_csv/nhgis0038_ds172_2010_place.csv	data/raw/census/	Used by <code>maxcitypop.do</code> for maximum principal-city population in 2010. TODO: Add NHGIS extract citation.
NHGIS extract	census/nhgis0041_csv/nhgis0041_csv/nhgis0041_ts_national_county.csv	data/raw/census/	Used by <code>cz_place_race_pop_1970_2010.do</code> for 1970 education variables. TODO: Add NHGIS extract citation.
NHGIS extract	census/nhgis0046_csv/nhgis0046_csv/nhgis0046_ts_national_county.csv	data/raw/census/	Used by <code>dataprep.do</code> for additional county/CZ controls. TODO: Confirm exact variables and extract citation.
NHGIS extract	census/nhgis0048_csv/nhgis0048_csv/nhgis0048_ds107_1980_county.csv	data/raw/census/	Used by <code>cz_pop_occscore_mfg.do</code> for 1980 manufacturing/labor-force shares. TODO: Add NHGIS extract citation.

Data source	Files or folders used by code	Expected location	Use and availability notes
NHGIS extract	census/nhgis0048_csv/nhgis0048_csv/nhgis0048_ds123_1990_county.csv	data/raw/census/	Used by <code>cz_pop_occscore_mfg.do</code> for 1990 manufacturing/labor-force shares. TODO: Add NHGIS extract citation.
NHGIS extract	census/nhgis0048_csv/nhgis0048_csv/nhgis0048_ds151_2000_county.csv	data/raw/census/	Used by <code>cz_pop_occscore_mfg.do</code> for 2000 manufacturing/labor-force shares. TODO: Add NHGIS extract citation.
NHGIS extract	census/nhgis0052_csv/nhgis0052_csv/nhgis0052_ds99_1970_place.csv	data/raw/census/	Used by <code>mechanisms.do</code> for 1970 white-flight / mover measures. TODO: Add NHGIS extract citation.
Census TIGER/Line place shapefiles	census/tiger/states/tl_2022_*_place.shp and sidecar files	data/raw/census/tiger/states/	Used by <code>municipal_shapefile.R</code> . TODO: Confirm all state shapefile sidecars and cite TIGER/Line 2022.
Census cartographic/TIGER cache	tigris-downloaded county/place shapes cached under <code>census/tiger/CB</code> , <code>census/tiger/full</code> , and <code>census/tiger/states_2011/</code> .	data/raw/census/tiger/ or downloaded by R package <code>tigris</code> .	Used by R spatial cleaning scripts. Requires network access if the <code>tigris</code> cache is not pre-populated. TODO: Confirm cache contents included in final deposit.
Census of Governments crosswalk	cog/4_Govt_Org_Directory_Surveys/GOVS_ID_to_FIPS_Place_Codes_2002.xls	data/raw/cog/	Used by <code>cog_cleaning.do</code> to build the COG ID to FIPS place crosswalk. TODO: Confirm citation and redistribution status.
Census of Governments crosswalk	cog/1-3_Govt_Org_Nat_CoArea_ElecOff/GOVS_to_FIPS_Codes_State_&County_2007.xls	data/raw/cog/	Used by <code>cog_cleaning.do</code> to crosswalk 2002/2007 county FIPS. TODO: Confirm citation.
Census of Governments crosswalk	cog/1-3_Govt_Org_Nat_CoArea_ElecOff/2012_GOVS_COUNTY_to_FIPS_COUNTY.xls	data/raw/cog/	Used by <code>cog_cleaning.do</code> to crosswalk 2012 county FIPS. TODO: Confirm citation.
Census of Governments county counts	cog/1-3_Govt_Org_Nat_CoArea_ElecOff/2_Govt_Org_County_Area_Counts.xls	data/raw/cog/	Used by <code>cog_cleaning.do</code> for municipal, county, school district, and special district counts. TODO: Confirm citation.
Census of Governments individual-government directory	ODBC DSN 4_Govt_Org_Directory_Surveys, pointing to the downloaded database/files under <code>cog/4_Govt_Org_Directory_Surveys/</code> .	data/raw/cog/ plus local ODBC DSN	Used by <code>cog_cleaning.do</code> for tables 2–4 of individual governments. TODO: Confirm exact database filename and whether it is included or must be downloaded.

Data source	Files or folders used by code	Expected location	Use and availability notes
Census Government Units file	cog/Govt_Units_2021_Final.xlsx	data/raw/cog/	Used by <code>cog_cleaning.do</code> for the 2021 master unit file. TODO: Confirm citation.
Census historical city finance database	ODBC DSN City_Gov_Fin, table City_Govt_Finances.	data/raw/cog/ or other local database path	Used by <code>cog_cleaning.do</code> for historical city-government finance variables. TODO: Confirm database filename, source, citation, and redistribution status.
Census Individual Finance file	census/_IndFin_1967-2012/IndFin12a.txt	data/raw/census/_IndFin_1967-2012/	Used by <code>IndFin_cleaning.do</code> to build <code>data/interim/census/IndFin12.dta</code> . TODO: The active master calls <code>code/cleaning/IndFin_cleaning.do</code> , but that exact file is missing.
Census Individual Finance file	census/_IndFin_1967-2012/IndFin12b.txt	data/raw/census/_IndFin_1967-2012/	Used by <code>IndFin_cleaning.do</code> . TODO: Confirm citation and redistribution status.
Census Individual Finance file	census/_IndFin_1967-2012/IndFin12c.txt	data/raw/census/_IndFin_1967-2012/	Used by <code>IndFin_cleaning.do</code> . TODO: Confirm citation and redistribution status.
NCES / CRDC / EDFacts / EDGE school data	nces/school-districts_lea_directory.csv; nces/EDGE_SCHOOLDISTRICT_TL23_SY2233/.../EDGE_SCHOOLDISTRICT_TL_23_SY2223.shp; pre-generated school_ccd_directory.dta, school_offerings.dta, school_race_ccd.dta, cz_achievement_segregation.dta.	data/raw/nces/; data/interim/nces/; data/xwalks/ncessch_place_xwalk.dta.	Used for school district geography, offerings, race/enrollment, court-order shares, and school segregation/achievement outcomes. TODO: The active code comments out the <code>educationdata</code> download commands, so the generated inputs and exact source steps need documentation.
School desegregation / court-order data	other/district_court_order_data_feb2021.dta	data/raw/other/	Used to construct commuting-zone court-order variables. TODO: Add creator, citation, access conditions, and redistribution status.
GNIS domestic names / streams	Files under GNIS/DomesticNames_AllStates_Text/Text/.	data/raw/GNIS/	Used to count stream features by commuting zone. TODO: Add exact GNIS download citation and date.
Transportation and railroad covariates	covariates/RR_NetworkDatabase_DH_Oct2015/.../Cost_ID_county.xlsx; NSFtranspCost.dta; Historical_Railroads___Vanderbilt.shp.	data/raw/covariates/	Used for transport cost and rail density controls. TODO: Add exact citations and redistribution terms.

Data source	Files or folders used by code	Expected location	Use and availability notes
Coastline and climate covariates	coastline-counties-list.xlsx; NOAA/NCEI climdiv-*.txt.	data/raw/covariates/	Used for coastal status and 1940 climate controls. TODO: Add exact citations and redistribution terms.
CoreLogic-derived place characteristics	corelogic/censusplace_clogic_chars.csv	data/interim/corelogic/	Used in municipal shapefile attributes and mechanisms. TODO: Confirm provenance, whether this can be redistributed, and whether only derived variables can be shared.
Generated and provided crosswalks	cw_cty_czone.dta; consistent_1990.dta; consistent_1940_1970.dta; place_county_xwalk.dta; cz_place_xwalk.dta; COG ID-FIPS crosswalks; ncessch_place_xwalk.dta.	data/xwalks/	Several are generated by the cleaning scripts, while others appear to be required inputs. TODO: Separate generated crosswalks from source crosswalks in the final deposit.

Dataset List

The code creates the following analysis and intermediate datasets. This list is based on `save`, `write_dta`, `write.csv`, and `st_write` commands in the programs called by `code/master.do`.

Dataset	Description	Created by
<code>data/clean/cz_pooled.dta</code>	Main commuting-zone analysis dataset; merges Derenoncourt city/CZ data, municipal outcomes, school district outcomes, controls, instruments, court-order variables, streams, population, income, education, and shift-share objects.	<code>code/cleaning/dataprep.do</code>
<code>data/clean/mechanisms.dta</code>	Place-level mechanisms dataset used for long-run mechanism, white-flight, distance, finance, land-use, and school exclusivity analyses.	<code>code/cleaning/mechanisms.do</code>
<code>data/clean/panel_a_data.dta</code>	Data for the paper's first-panel migration/instrument figure.	<code>code/cleaning/panel_a_data.do</code>
<code>data/clean/pccarrow_fig_data.dta</code>	Data for principal-city arrow figures.	<code>code/cleaning/pccarrow_fig_data.do</code>
<code>data/interim/dcourt/*.dta</code>	Derenoncourt-derived city/CZ datasets, original 130 CZ sample, white migration LASSO prediction files, and city-CZ crosswalks.	<code>code/dcourt_setup/*.do</code> ; <code>code/cleaning/2_lasso_white.do</code>
<code>data/interim/ssaggregate_prep/*.dta</code>	Shift-share shares, shocks, destination instruments, interacted shares, and 1,000 placebo instrument files.	<code>code/cleaning/create_sumshares.do</code> ; <code>code/ado/create_sumshare*.ado</code>
<code>data/interim/counts/*_cz.dta</code>	CZ-level municipal, county, school district, and special-purpose government counts.	<code>code/cleaning/muni_counts.do</code> ; <code>code/cleaning/cog_cleaning.do</code>
<code>data/interim/census/*.dta</code>	Income, education, population, occupation score, manufacturing share, urban population, municipal finance, and linked migrant characteristics.	Multiple cleaning scripts
<code>data/interim/cgoodman/*.dta</code>	C. Goodman place-county geography, original geographies, and CZ geography summaries.	<code>code/cleaning/cgoodman_place_county_geog.R</code> ; <code>code/cleaning/geogs.do</code>
<code>data/interim/cog/*.dta</code>	Census of Governments county counts, individual-government extracts, 2021 master unit file, and historical city finance data.	<code>code/cleaning/cog_cleaning.do</code>
<code>data/interim/other/*.dta, *.csv, *.shp</code>	Municipal shapefile and attributes, spatial covariates, central-city distances, covered exclusive school district indicator, streams.	R and Stata cleaning scripts

Dataset	Description	Created by
<code>data/interim/nces/*.dta</code>	School offerings, place offerings, CZ court order variables, and school achievement/segregation files.	<code>code/cleaning/ncessch_cleaning.do</code> ; TODO: achievement/segregation source script not in active master
<code>data/xwalks/*.dta</code>	Crosswalks used in cleaning and analysis.	Some generated by current cleaning scripts; some required as inputs

Computational Requirements

Software Requirements

The main code is Stata and R. `code/master.do` declares `version17.0`, so Stata 17 or newer is recommended. The master file currently points to R 4.4.2 on one machine and R 4.3.1 on another via `$Rterm_path`. The code has hard-coded Windows paths; no Linux/macOS run was verified while preparing this README draft.

- Stata 17 or newer.
- R, currently configured for R 4.4.2 or R 4.3.1 depending on username.
- Stata community packages can be installed by running `docode/setup_stata_packages.do` from the repository root. The setup script installs the Stata packages used by the active project code and by the edited Derenoncourt replication scripts, including `rsource`, `statastates`, `estout`, `ivreg2`, `ranktest`, `reghdfe`, `ftools`, `parmest`, `unique`, `labutil`, and related plotting/table utilities.
- R packages used by scripts called from `master.do`: `tidyverse`, `sf`, `haven`, `tigris`, `stringr`, `readxl`, `terra`, and `glmnet`. Some scripts install missing R packages from CRAN, which requires internet access.
- The NCES pre-generation comments use the Stata package `educationdata`. If those commented downloads are reactivated, document the package version and source queries.

TODO: Create an R package lockfile or setup script. The AEA guidance recommends not assuming that the replicator already has user-written packages installed.

ODBC Setup for Census of Governments

`code/cleaning/cog_cleaning.do` uses Stata's `odbc` commands for two Census of Governments inputs. The machine running the replication must have an ODBC driver that matches the Stata architecture. For 64-bit Stata on Windows, install the 64-bit Microsoft Access Database Engine / Access ODBC driver if it is not already available, then open the 64-bit ODBC Data Source Administrator at `C:/Windows/System32/odbcad32.exe`. For 32-bit Stata, use the 32-bit administrator at `C:/Windows/SysWOW64/odbcad32.exe`.

Create either User DSNs or System DSNs with these exact names:

- `4_Govt_Org_Directory_Surveys`: point this DSN to the downloaded Census of Governments individual-government directory database in `$RAWDATA/cog/4_Govt_Org_Directory_Surveys/`. The script queries this DSN, reads tables 2–4, and writes `data/interim/cog/4_*.dta`.
- `City_Gov_Fin`: point this DSN to the historical city-government finance database containing table `City_Govt_Finances`. The script writes `data/interim/cog/city_gov_fin.dta`.

After creating the DSNs, test them in Stata before running the full master file:

```
odbc list
odbc query "4_Govt_Org_Directory_Surveys"
odbc query "City_Gov_Fin"
```

If Stata cannot see either DSN, confirm that the DSN was created in the administrator matching the Stata bitness and that the database driver can open the downloaded file. Stata's Windows ODBC FAQ is linked in `code/cleaning/cog_cleaning.do`: <https://www.stata.com/support/faqs/data-management/configuring-odbc-win/>.

TODO: Confirm the exact database filenames for both DSNs and whether they will be redistributed in the replication package or must be downloaded by replicators.

Controlled Randomness

The active code does not set a Stata seed. The file `code/cleaning/2_lasso_white.do` calls R's `cv.glmnet` with `nfolds=5` and does not set an R seed in the embedded R block. The code also uses 1,000 Derenoncourt placebo files as inputs generated by the Derenoncourt replication workflow.

TODO: Set and document an R seed before each `cv.glmnet` call, or document why the resulting selected variables are fixed by provided intermediate outputs. Confirm whether the Derenoncourt placebo files were generated with a documented seed.

Memory, Runtime, and Storage Requirements

TODO: Add the machine specifications, operating system, memory, runtime, and storage requirements from a clean full run of `code/master.do`.

Approximate runtime category:

- ☐ Less than 10 minutes
- ☐ 10–60 minutes
- ☐ 1–2 hours
- ☐ 2–8 hours
- ☐ 8–24 hours
- ☐ 1–3 days
- ☐ 3–14 days
- ☐ More than 14 days

Approximate storage requirement:

- ☐ Less than 25 MB
- ☐ 25 MB–250 MB
- ☐ 250 MB–2 GB
- ☐ 2 GB–25 GB
- ☐ 25 GB–250 GB
- ☐ More than 250 GB

Description of Programs and Code

`code/master.do` is the master script. It performs the following steps:

1. Sets user-specific paths, Stata version, Stata globals, and the ado path.
2. Writes `paths.csv`, which several R scripts read to recover Stata global paths.
3. Copies edited Derenoncourt setup scripts into the Derenoncourt replication directory via `code/cleaning/dcourt_file_setup.do`.
4. Runs the edited Derenoncourt replication master file at `$RAWDATA/dcourt/replication_AE R/code/0_MASTER_edited.do`.
5. Runs cleaning scripts under `code/cleaning`.
6. Runs analysis scripts under `code/analysis`.

Cleaning Code

The cleaning stage constructs crosswalks, Derenoncourt-derived inputs, municipal counts, Census/IPUMS/NHGIS-derived controls, spatial covariates, school and court-order inputs, and the final clean analysis datasets. Key outputs are `data/clean/cz_pooled.dta`, `data/clean/mechanisms.dta`, `data/clean/panel_a_data.dta`, and `data/clean/pccarrow_fig_data.dta`.

Analysis Code

The analysis stage reads the cleaned datasets and writes manuscript and appendix outputs to `exhibits/tables` and `exhibits/figures`. Stata helper programs in `code/ado` produce repeated table formats and shift-share/segregation calculations.

Known Execution Issues to Resolve Before Deposit

- `code/master.do` calls `code/cleaning/migrate_cleaning.do`, but that exact file is not present. A likely related file exists at `code/cleaning/old/migrate_cleaning.do`. **TODO:** Confirm whether to restore it, remove the call, or replace it with the Derenoncourt-generated migration input.
- `code/master.do` calls `code/cleaning/create_sumshare.do`, but the repository contains `code/cleaning/create_sumshares.do`. **TODO:** Confirm whether this is a typo in the master script.
- `code/master.do` calls `code/cleaning/IndFin_cleaning.do`, but that exact file is not present. A likely related file exists at `code/cleaning/old/IndFin_cleaning.do`. **TODO:** Confirm whether to restore it.
- `code/cleaning/ncessch_cleaning.do` assumes `data/interim/nces/school_ccd_directory.dta`, `school_offerings.dta`, and `school_race_ccd.dta` already exist; the download commands are commented out. **TODO:** Document and automate creation of these files or include them as inputs with source details.
- `code/analysis/segregation_table.do` uses `data/interim/nces/cz_achievement_segregation.dta`, but the active master does not generate it. A related script exists under `code/cleaning/old/ap_de_cleaning.do`. **TODO:** Confirm source and include active generation code if needed.
- `code/cleaning/muni_district_overlap.R` uses `INTDATA` and `RAWDATA` directly and does not read `paths.csv`, unlike the other R cleaning scripts. **TODO:** Confirm that these objects are defined under `rsource`; otherwise add the same path-loading block used by the other R scripts.

License for Code

TODO: Specify the code license. The repository includes `LICENSE`; confirm whether the license is intended for all code in the replication package and whether any bundled third-party code or ado files require separate notices.

Instructions to Replicators

1. Obtain the data sources listed in the data availability section and place them in the directory structure expected by `code/master.do`. By default, the current local path is `D:/munis_replication/data`.

2. Edit the path block near the top of `code/master.do` so that `$DROPBOX`, `$REPO`, `$Rterm_path`, and `$FFMPEG` point to local paths on the replicator's machine.
3. In Stata, from the repository root, run `docode/setup_stata_packages.do` to install required user-written Stata packages.
4. Install required R packages, or allow the R scripts that install missing packages to use internet access. **TODO:** Replace this with an R setup script or lockfile once created.
5. Configure the two Census of Governments ODBC DSNs described above if running `code/cleaning/cog_cleaning.do`.
6. Ensure that `data/raw/dcourt/replication_AER/` contains the Derenoncourt replication package, because `master.do` copies edited scripts into that directory and then runs `O_MASTER_edited.do`.
7. Resolve the known execution issues listed above, especially the three missing master-called files.
8. In Stata, from the repository root, run `docode/master.do`.
9. Outputs will be written to `exhibits/tables` and `exhibits/figures`.

List of Tables, Figures, and Programs

The provided code is intended to reproduce:

- ☐ All numbers provided in text in the paper.
- ☐ All tables and figures in the paper.
- ☐ Selected tables and figures in the paper, as explained below.

TODO: Check the appropriate box after comparing this output list with the final manuscript and appendix.

Figure/Table	Program	Output file(s)	Notes
Table 1	code/analysis/summary_table.do	exhibits/tables/T1.tex	Summary statistics.
Table 2	code/analysis/main_table.do; main_table_long_ssaggregate.ado	exhibits/tables/T2.tex	Main 2SLS table.
Table 3	code/analysis/main_table.do; fragmech_long_ssaggregate.ado	exhibits/tables/T3.tex	Fragmentation/mechanism table.
Table 4	code/analysis/occscore_links_diffs_ss.do	exhibits/tables/T4.tex	Linked OCCSCORE heterogeneity. TODO: Confirm manuscript title.
Table 5	code/analysis/long_term_mechanisms.do	exhibits/tables/T5.tex	Long-term mechanisms.
Table 6	code/analysis/segregation_table.do	exhibits/tables/T6.tex	Long-term segregation and achievement-related outcomes.
Figure 1, panels D–E	code/analysis/fig_1_panels_b_c.do	exhibits/figures/F1d.pdf; F1e.pdf	TODO: Confirm panel labels; program name says panels B/C.
Figure 2, panels A–B	code/analysis/event_studies.do	exhibits/figures/F2a.png; F2b.png	Event studies.
Figure 3, panels A–B	code/analysis/pcarrow_fig.do	exhibits/figures/F3a.pdf; F3b.pdf	Principal-city arrow figures.
Appendix Table A1	code/analysis/balancetables.do	exhibits/tables/TA1.tex	Balance table for baseline instrument.
Appendix Table A2	code/analysis/pretrends.do	exhibits/tables/TA2.tex	Pre-trends table.
Appendix Table A3	code/analysis/main_table.do	exhibits/tables/TA3.tex	Town outcome robustness.
Appendix Table A4	code/analysis/corr_matrix.do	exhibits/tables/TA4.tex	Correlation matrix.
Appendix Tables A5–A7	code/analysis/main_table.do	TA5.tex; TA6.tex; TA7.tex	Main-table robustness variants.
Appendix Tables A8–A12	code/analysis/imbalance_decomp.do	TA8.tex–TA12.tex	Imbalanced-controls decomposition for <code>cgoodman</code> , <code>gen_muni</code> , <code>schdist_ind</code> , <code>spdist</code> , and <code>totfrac</code> .

Figure/Table	Program	Output file(s)	Notes
Appendix Tables A13–A14	code/analysis/main_table.do	TA13.tex; TA14.tex	Control-set robustness.
Appendix Table A15 and auxiliary files	code/analysis/balancecontrols_individualeffects.do	TA15.tex; balancecontrols_rf.tex; balancecontrols_rf_sep.tex	Baseline controls individual effects and auxiliary reduced-form balance files. TODO: Confirm whether auxiliary files appear in manuscript/appendix.
Appendix Table A16	code/analysis/balancetables.do	exhibits/tables/TA16.tex	Percentile instrument balance table.
Appendix Table A17	code/analysis/main_table.do	exhibits/tables/TA17.tex	Non-ssaggregate / alternative instrument version. TODO: Confirm description.
Appendix Table A18	code/analysis/main_table.do	exhibits/tables/TA18.tex	Weighted instrument-migration robustness.
Appendix Table A19	code/analysis/balancetables.do	exhibits/tables/TA19.tex	White migration balance table.
Appendix Table A20	code/analysis/main_table.do	exhibits/tables/TA20.tex	White migration robustness table.
Appendix Tables A21–A22	code/analysis/occscore_links_diffs_fs.do	TA21.tex; TA22.tex	First-stage linked OCCSCORE heterogeneity by links/differences.
Appendix Table A23	code/analysis/touching_pct_rev_debt_table.do	exhibits/tables/TA23.tex	Boundary/revenue/debt mechanisms.
Appendix Table A24	code/analysis/main_court_ordered.do	exhibits/tables/TA24.tex	Court-ordered heterogeneity.
Appendix Table A25	code/analysis/white_flight_check.do	exhibits/tables/TA25.tex	White-flight check.
Appendix Figure A1	code/analysis/municipal_incorporations_graph.do	exhibits/figures/FA1.png	Municipal incorporations over time.

Figure/Table	Program	Output file(s)	Notes
Appendix Figure A2 series	<code>code/analysis/loo_test.do</code>	FA2_<outcome>.pdf	Leave-one-out tests for <code>cgoodman</code> , <code>gen_muni</code> , <code>gen_town</code> , <code>schdist_ind</code> , <code>spdist</code> , <code>totfrac</code> .
Appendix Figure A3 series	<code>code/analysis/alt_inst_tests.do</code>	FA3_<outcome>.pdf	Alternative instrument tests for the same outcome family.
Appendix Figure A4 series	<code>code/analysis/placebo_test.do</code>	FA4_<outcome>.pdf	Derenoncourt placebo tests for the same outcome family.
Appendix Figures A5a–A5b	<code>code/analysis/dist_edge_edge.do</code>	FA5a.pdf; FA5b.pdf	Distance to principal city graphs.

References

TODO: Replace this draft reference list with final journal-formatted citations and ensure that all data citations also appear in the manuscript or appendix if required.

- Derenoncourt, Ellora. 2022. **TODO:** Add full article title, journal, replication package URL/DOI, and access date.
- Goodman, Christopher B. **TODO:** Add full citation for municipal incorporation data.
- Manson, Steven, Jonathan Schroeder, David Van Riper, Tracy Kugler, and Steven Ruggles. IPUMS National Historical Geographic Information System: Version used in this project. Minneapolis, MN: IPUMS. **TODO:** Add extract-specific citation/DOI and access date.
- Ruggles, Steven, Sarah Flood, Matthew Sobek, Daniel Backman, Annie Chen, Grace Cooper, Stephanie Richards, Renae Rodgers, and Megan Schouweiler. IPUMS USA: Version used in this project. Minneapolis, MN: IPUMS. **TODO:** Add exact version/DOI and extract access date.
- U.S. Census Bureau. Census of Governments, Government Organization and Individual Unit / Individual Finance files. **TODO:** Add exact file citations and dates.
- U.S. Census Bureau. TIGER/Line Shapefiles and Cartographic Boundary Files. **TODO:** Add exact years and access date.
- National Center for Education Statistics and Civil Rights Data Collection / EDData. **TODO:** Add exact dataset citations and access dates.
- U.S. Board on Geographic Names. Geographic Names Information System. **TODO:** Add exact download citation and access date.
- National Centers for Environmental Information. Climate Division monthly data. **TODO:** Add exact citation and access date.
- **TODO:** Add citations for transportation-cost data, historical railroad shapefile, coastline county list, CoreLogic-derived data, court-order data, Geocorr crosswalks, and all source crosswalks.

Acknowledgements

This README structure follows the social science replication README template and the AEA Data Editor guidance for preparing files for verification. See <https://aeadataeditor.github.io/aea-de-guidance/preparing-for-data-deposit.html>.